

SPECIFICATION

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1 FOREWORD

This document is intended for educational purposes only and shall not be used for construction or contractual purposes.

2 GENERAL

2.1 General information

This specification, with the attached general arrangement drawing NI24.1133_N-000-001-GENERAL ARRANGEMENT, describes the design, materials, construction and equipment of a research vessel.

2.2 Definitions

The customer is a person or entities who, after the completion of the construction of the vessel and carrying out sea trials, will receive it and will be its owner.

Supplier - the builder (shipyard) that carries out the construction of the vessel, unless otherwise specified (for example, Paint Supplier)

Classification Society - An organization under whose supervision and according to the rules a ship is designed and built, and which, after sea trials, issues all the documents necessary for its normal operation (certificates, certificates, acts, etc.)

2.3 Systems of measures and units of measurement

This specification uses the metric system of measures.

Name	Units	Designation
Length	meter/mile	m, mi 1 mile = 1852 m
Area	square meter	m ²
Volume	cubic meter	m ³
Mass	kilogram/tonne	Kg 1 t = 1000 kg
Density	tons per cubic meter	t/m ³
Time	minute	sec 1 min = 60 sec
Force	newton	N = kg*m/sec ²
Energy	joule	J = N*m
Power	watt/kilowatt/horse power	1 hp = 736 W 1 W = 1J/sec 1 kW = 1000 W
Electrical power	volt-ampere	VA
Current intensity	ampere	A
Frequency	hertz/revolutions per minute	Hz/rpm
Temperature	degrees Celsius	°C

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Name	Units	Designation
Pressure	pascal/kilopascal/bar	1 Pa = 1 N/m ² 1 kPa = 1000 Pa 1 Bar = 100 kPa

2.4 Environmental characteristics

Density of:	
sea water, t/m ³	1.025
fresh water, t/m ³	1.000
fuel oil, t/m ³	0.860
lubrication oil, t/m ³	0.900
Temperature of water:	
minimum	0 °C
maximum	+30 °C
Temperature of air:	
minimum	-20 °C
maximum	+35 °C

2.5 Type of the vessel

The “Gunnerus” is a multipurpose research vessel owned and operated by the Norwegian University of Science and Technology (NTNU), Faculty of Natural Sciences and Technology.

Designed by	Polarkonsult AS in Harstad, Norway
Built by	Yard no. 41 at Larsnes Mekaniske Verksted, Norway, in 2006.
Conversions	2015: Exchanging shaft propellers, nozzles and rudders for twin ducted azimuth thrusters. 2019: Elongated by 5 metres to increase the vessel capacities.
Port of registry	Trondheim, Norway
Call Sign	LNVZ
IMO no.	9371361
Classification	Norwegian Maritime Authorities (NMA)

2.6 Navigation areas and operating conditions

Built for coastal areas within 20 nautical miles from the coast (Liten Kystfart).

Designed and built according to European trade.

Built according to class notation: DNV +1A1 Ice-C, E0, R2, Cargo ship.

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2.7 Main dimensions

Length over all, m	36.25
Length between p.p., m	33.90
Length at waterline, m	34.90
Breadth moulded, m	9.60
Breadth extreme, m	9.90
Depth to 1-deck, m	4.29
Draught underside keel at stern, m	3.70
Baseline draught amidships, m	2.78
Mast height / antenna, m	17.40/22.25
Gross Tonnage, GT	430
Net Tonnage, NT	130

2.8 Capacities

Accommodation	23 bunks in 8 cabins
Daytime capacity	25 incl. crew (4-5 crew)
Deadweight	169 t
Deck load	38 t
Fuel oil	60 m ³
Fresh water, potable	18 m ³
Fresh water, technical	62 m ³
Cargo hold	38 m ³
Messroom	21 chairs and 46" LCD monitor
Conference room	8 chairs, network and no monitor

2.9 General arrangement

The vessel has a main deck with the diving platform, A-frame, and a work boat on it. A forecastle with accommodation and a forepeak store. Superstructure deck is fitted with lifesaving, anchoring, and mooring equipment. Superstructure is fitted with the ventilation and electronics room, galley, and a day room.

The ship is divided into 6 compartments by 5 transverse watertight bulkheads.

2.9.1 Arrangement of compartments and cabins

Four transverse watertight bulkheads at 8, A, 32, and 51 fr. divide the ship's hull into five watertight compartments.

2.9.1.1 Compartments

Thruster room 0...3 fr.

Cargo hold 3...8 fr.

Engine room 8...A fr.

Accommodation A...32 fr.

Accommodation 32...51 fr.

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Fore peak compartment 51...59

2.9.1.2 Tanks

No 1 Dbb water ballast,	48...51 fr.
No 3sb Dbb water ballast,	32...40 fr.
No 3p Dbb water ballast,	32...40 fr.
No 12 Roll dumping tank,	8...11 fr.
No 13sb Wing tank water ballast,	2...8 fr.
No 13p Wing tank water ballast,	2...8 fr.
No 5 Dbb fresh water,	32...37 fr.
No 2sb Dbb fuel oil,	40...48 fr.
No 2p Dbb fuel oil,	40...48 fr.
No 6sb Dbb fuel oil settling,	20...28 fr.
No 6p Dbb fuel oil,	20...28 fr.
No 11sb FO Service tank,	17...20 fr.
No 11p FO Service tank,	16...19 fr.
New fuel oil tank,	A...I fr.
New fuel oil tank,	A...I fr.
No 7sb Black sewage,	22...26 fr.
No 7p Gray sewage,	22...26 fr.
Black water new,	A...E fr.
Gray water new,	A...E fr.
No 8 Sludge,	20...22 fr.
No 9 Waste,	18...20 fr.
No 10 Bilge,	18...22 fr.
No 4 Dbb dry tank,	37...40 fr.
Hold Cargo hold aft,	2...8 fr.

2.10 Vessel speed

Speed at 100% MCR	11.9 knots
Cruising speed	9.5 knots

2.11 Stability

The stability of the ship is ensured in all cases of loading, according to the DNV rules.

2.12 Class, Rules and Conventions

The ship is designed according to BV Rules for the class:

DNV +1A1 Ice-C, E0, R2, Cargo ship

The ship complies with the current conventions and rules:

- DNV-RU-SHIP;
- Norwegian Maritime Authorities (NMA);
- “International Convention on Load Lines” (CLL – 66) edition 1966;
- “International Convention on Tonnage Measurement of Ships”, 1969

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3 HULL

3.1 General information

The design, materials, and strength correspond to the purpose and specified conditions of navigation and operation of the vessel and comply with the applicable rules, regulations, and conventions.

The ship hull is designed for ice class ICE C. The ship hull, superstructure, and wheelhouse are steel; all elements are welded.

Dimensions of structural components of the ships hull correspond to the requirements of DNV Rules.

3.1.1 Frame-spacing

Transverse frames of ship hull, superstructure and wheelhouse throughout the length is 500 mm.

3.2 Main hull

The ship hull is designed according to the transverse framing system. The dimensions of the structure elements of the hull comply with the DNV Rules. The vessel is divided into 7 compartments by transverse watertight bulkheads. Double bottom is located on the 18...51 frames.

3.3 Wheelhouse and superstructure

The superstructure is made according to the transverse framing system. The superstructure breadth is 7000 mm.

The wheelhouse is located in the middle of the ship, and has windows for all-around visibility.

4 SHIP GEARS

4.1 Propulsion complex

The propulsion system on the ship is made of two 500 kW Azipods.

Propeller diameter - 1900 mm

To ensure maneuverability, the vessel is equipped with a 200 kW bow thruster with a diameter of 800 mm.

4.2 Anchor gear

4.2.1 Anchors

The vessel is equipped with two bow anchors HHP. The weight of the anchors is 495 kg each.

4.2.2 Winch

For raising and releasing anchors, the vessel is provided with a winch with 2 drums (20 m/min, 2 x 12,5m Ø22mm K2 chain/ 210m Ø22mm wire).

4.2.3 Anchor hawsepipes

Two anchor hawsepipes ensure stowed position and swinging-out of anchors overboard when releasing. Washing by water from water fire extinguishing system is provided for anchors and anchor chains.

4.3 Mooring equipment

4.3.1 Bollards

Bollards are provided in quantity 8 to perform mooring operations on the ship.

4.4 Deck Equipment, Scientific Equipment and Lab Facilities

Trawl winches	2 x 6 t, (each with 1000 m wire Ø14 mm)
Net drum	5 m ³ , shields 2000mm, core 320mm
Main deck crane	18 m / 62 tm
CTD crane	5 m / 3.3 tm, Water sampler wire 5 mm/750m CTD wire 6.5 mm/750 m
Stern mounted A-frame	6 t, hydraulic
Side monted A-frame	4 t SWL, hydraulic
ROV hangar / skidding plate	
Hydraulic diving platform	500 kg, 1.5m x 0.8m
Hydraulic aggregate	110 kW
Capstan	8t/220bar, D=410, d=320, L=300
Anchor winch	2 drums, 20 m/min, 2 x 12,5m Ø22mm K2 chain/ 210m Ø22mm wire
Compressed air	
Working deck	110 m ²

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Wet lab	14 m ²
Dry lab	12 m ²
Computer lab	12 m ²
Container capacity	5, 10, 15 & 20 feet alongside or 20 feet abeam
CTD	
Water sampling system	
Workboat	80 hp

4.5 Collective and personal lifesaving appliances

MOB boat	6 persons, SOLAS approved, outboard 20 hp
Rescue boat davit	
Life rafts	2 x 25 man
Survival suits	25 pcs
Work vests	
Life jackets	25 pcs
Lifebuoy	8 pcs
EPIRB	
SART	
Aircraft beacon	
Fire alarm system	
Fixed fire-fighting system	CO2 system in engine room
Fire suits	
Search light	

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5 PROPULSION PLANT

5.1 General

Each generator set (diesel engine and generator) is mounted on a common double elastic frame and one of the sets is mounted in a noise-dampening hood for a special low noise mode.

5.2 Main propulsion plant

The main propulsion plant includes three diesel engines with a power of 450 kW each and three generators.

6 ELECTRICAL EQUIPMENT

Battery Box	
Transformer	3x440/230 Volt
Main Switchboard	
Frequency Converter	
Frequency Converter Bowthruster	
El. Starter Boxes	
El. Cabinets	

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7 NAVIGATION, COMMUNICATIONS AND ELECTRONIC EQUIPMENT

Dynamic Positioning system	
DP - Reference systems	Differential GPS Acoustic positioning Relative position reference
Heading, Attitude and Positioning Sensor	GNSS-based heading, attitude and position sensor
Acoustic positioning system	Operating range up to ~5000 m
Motion reference unit (MRU)	
Autopilot	
Compass, magnet	
Compass, gyro	
Bearing repeater	
Differential positioning sensor	
Heading, attitude and positioning sensor	
GPS	10 m position accuracy (95%) and 6" LCD
Radar	X-band radar, range up to 48 nm
Radar Transponder	
Log	0–60 knots, depth 0.5–200 m
Echo sounder	38, 50, 200 kHz – 2000m
Echo sounder	300 kHz, depth range 0–200 m, swath up to 140°
Catch monitoring system	Up to 6 sensors
Chartplotter 1	Electronic chart system with AIS
Chartplotter 2	Electronic chart system with AIS, sonar, multibeam, hydrographic tools
Chartplotter 3	Office version of Olex for chart analysis and planning
AIS Furuno	AIS Class A, transmit 12.5 W, receive 1 W, range up to ~20 nm
Navtex	Navtex receiver, frequency 490/518 kHz, reception range up to ~400 NM
GMDSS console	
VHF fixed radio	
VHF handheld radios	
UHF handheld radios	
Satellite phone	
Internet at sea	
Internet in port in Trondheim	Wireless broadband NTNU

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